03 Probability

1. INTRODUCTION:

The theory of probability has been originated from the game of chance and gambling. In old days, gamblers used to gamble in a gambling house with a die to win the amount fixed among themselves. They were always desirous to get the prescribed number on the upper face of a die when it was thrown on a board. Shakuni of Mahabharat was perhaps one of them. People started to study the subject of probability from the middle of seventeenth century. The mathematicians Huygens, Pascal, Fermat and Bernoulli contributed a lot to this branch of Mathematics. A.N. Kolmogorow proposed the set theoretic model to the theory of probability.

Probability gives us a measure of likelihood that something will happen.

However probability can never predict the number of times that an occurrence actually happens. But being able to quantify the likely occurrence of an event is important because most of the decisions that affect our daily lives are based on likelihoods and not on absolute certainties.

2. DEFINITIONS:

2.1 Experiment:

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In probability theory, an experiment (or trial) is any procedure that can be infinitely repeated and has a well-defined set of possible outcomes.

2.2 Random Experiment:

An experiment is said to be random if it has more than one possible outcome. Before rolling a die we do not know the outcome, but we do know all the possible outcomes. Only after performing the experiment, it's outcome is known. This is an example of a random experiment. In particular, a random experiment is an experiment whose all possible outcomes are well defined but cannot be predicted before actually performing the experiment.

Other examples of random experiment include tossing a coin, choosing a natural number less than 10, drawing a card from a pack of well shuffled playing cards etc.

2.3 Sample Space:

A set that consists of all possible outcomes of a random experiment is called a sample space and each outcome is called a sample point. Often, there will be more than one sample space that can describe outcomes of an experiment, but there is usually only one that will provide the most information.

For e.g. in an experiment of "throwing a die", following sample spaces are possible :

(i) S_1 = {even number, odd number}

(ii) S_2 = {a number less than 3, a number equal to 3, a number greater than 3}

(iii) S_3 = {1,2,3,4,5,6}

Here sample space S_3 is the one which provides most information.

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2.4 Finite and Infinite Sample Space:

If a sample space has a finite number of points it is called finite sample space and if it has an infinite number of points, it is called infinite sample space.

For e.g.

(i) In random experiment "toss of coin" either a head (H) or tail (T) comes up, therefore sample space of this experiment is $S = \{H, T\}$ which is a finite sample space.

(ii) Sample space of experiment "Selecting a number from the set of natural numbers, is $S = \{1,2,3,4,...\}$ which is an example of infinite sample space.

2.5 Event:

Any subset of sample space is called an event. For e.g.

(i) Getting a head in a toss of a coin.

(ii) Scoring a six on the throw of a die,

(iii) Winning the first prize in a raffle,

Note:

(i) An event consisting of a single point of S is called a **simple** or **elementary** event.
For e.g. Getting a Heads in toss of a coin.

(ii) An event consisting of 0 outcomes is called **impossible** event or **null** event.
For e.g. Getting a sum of 13 in random experiment of "rolling of two dice".

(iii) An event consisting of all possible outcomes is called **sure** event.
For e.g. Getting a natural number in random experiment of "rolling of a die".

2.6 Compound Event:

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If an event has more than one sample points, it is called a **Mixed or Compound Event**.

For e.g.

(i) Getting a sum of 10 in random experiment of "rolling of two dice".

(ii) Getting both Heads and Tails in random experiment of "toss of two coins".

2.7 Complement of an event:

The set of all outcomes which are in S but not in A is called the complement of the event A & denoted by $\bar{A}$, A^c, A' or 'not A'.

For e.g.

(i) Event 'Getting at least one Heads' is complement of event 'Getting 3 Tails' in random experiment of "toss of three coins".

(ii) In random experiment of "rolling of two dice", Event 'getting a sum greater than 7' is complement of event 'getting a sum less than 8'.

2.8 Mutually Exclusive Events:

Two events are said to be **mutually exclusive** (or disjoint or incompatible) if the occurrence of one precludes (rules out) the simultaneous occurrence of the other.

Note:

(i) If A and B are two given events then event $A \cap B$ is also denoted by AB or A & B.

(ii) If A and B are two mutually exclusive events then $P(A \cap B) = 0$.

(iii) If A, B and C are mutually exclusive events then $P(A \cap B) = P(B \cap C) = P(C \cap A) = 0$.

(iv) If events A_i; $i = 1, 2, 3, ..., n$ are mutually exclusive then $P(A_i \cap A_j) = 0$ for all $i \neq j$.

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Example:

Consider, a random experiment of choosing a number from the set {3, 4, 5, 6, 7, 8, 9, 10, 11, 12}.

If, Event A is the selection of a prime number,

Event B is the selection of an odd number,

Event C is the selection of an even number,

then

A, C are mutually exclusive as none of the numbers in this set is both prime and even, and A, B are not mutually exclusive as few numbers are both prime and odd (viz. 3, 5, 7, 11).

2.9 Equally Likely Events:

Events are said to be **equally likely** when each event is as likely to occur as any other event. Note that the term 'at random' or 'randomly' means that all possibilities are equally likely.

Note :

(i) If A and B are two equally likely events then $P(A) = P(B)$.

(ii) If A, B and C are equally likely events then $P(A) = P(B) = P(C)$.

(iii) If events A_i; $i = 1, 2, 3, ..., n$ are equally likely events then $P(A_i) = P(A_j)$ for all $i \neq j$.

2.10 Exhaustive Events:

A set of evesnts are said to be **Exhaustive Events** if no event outside this set can result as an outcome of an experiment.

Note:

(i) If A and B are exhaustive events then $P(A \cup B) = 1$.

(ii) If A, B and C are exhaustive events then $P(A \cup B \cup C) = 1$.

(iii) If events A_i; $i = 1, 2, 3, ..., n$ are exhaustive events then $P(A_1 \cup A_2 \cup ... \cup A_n) = 1$.

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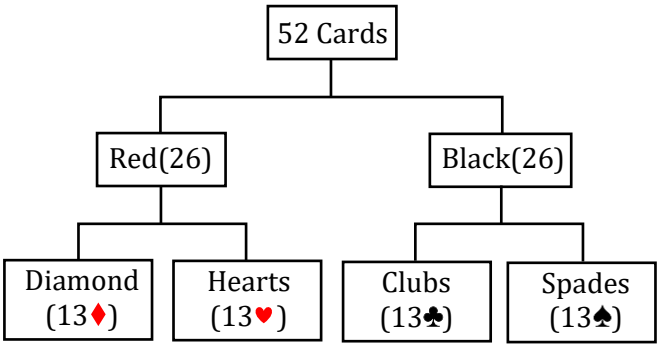
2.11 Comparative study of Equally likely, Mutually Exclusive and Exhaustive events:

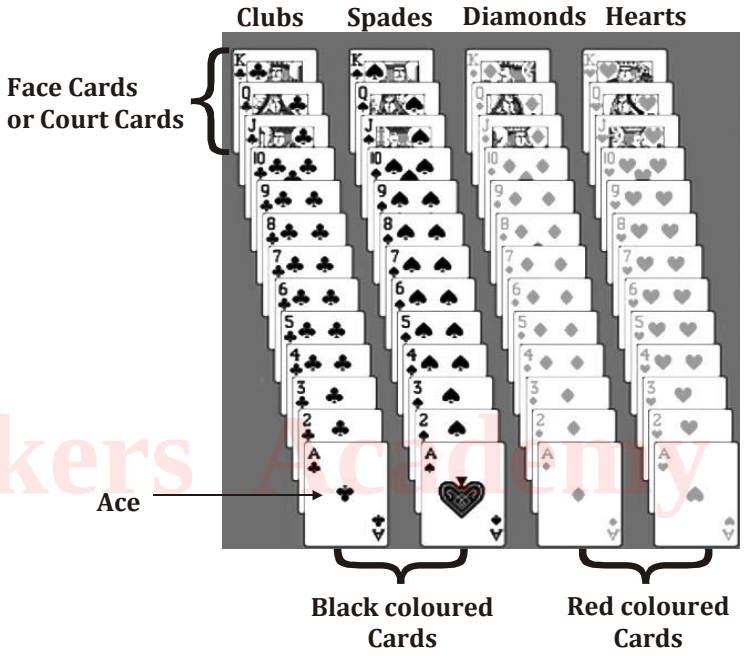
	Experiment	Events	E/L	M/E	Exhaustive
1.	Throwing of a die	A: throwing an odd face {1, 3, 5} B : throwing a composite {4,6}	No	Yes	No
2.	A ball is drawn from an urn containing 2White, 3Red and 4Green balls	E_1 : getting a White ball E_2 : getting a Red ball E_3 : getting a Green ball	No	Yes	Yes
3.	Throwing a pair of dice	A : throwing a doublet {11, 22, 33, 44, 55, 66}			
		B : throwing a total of 10 or more {46, 64, 55, 56, 65, 66}	Yes	No	No
4.	From a well shuffled pack of cards a card is drawn	E_1 : getting a heart E_2 : getting a spade E_3 : getting a diamond E_4 : getting a club	Yes	Yes	Yes
5.	From a well shuffled pack of cards a card is drawn	A = getting a heart B = getting a face card	No	No	No

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2.12 Playing cards:

A pack of playing cards consists of 52 cards of 4 suits, 13 in each, as shown in figure.





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Illustration 1

Write the sample space of the following experiments:

A : Rolling two dice

B : Tossing two coins

C : A coin tossed thrice

D : Draw a card from a well shuffled pack and note its suit.

Solution:

$$S_A = \begin{Bmatrix} (1,1),(1,2),(1,3),(1,4),(1,5),(1,6), \\ (2,1),(2,2),(2,3),(2,4),(2,5),(2,6), \\ (3,1),(3,2),(3,3),(3,4),(3,5),(3,6), \\ (4,1),(4,2),(4,3),(4,4),(4,5),(4,6), \\ (5,1),(5,2),(5,3),(5,4),(5,5),(5,6), \\ (6,1),(6,2),(6,3),(6,4),(6,5),(6,6) \end{Bmatrix}$$

$$S_B = \{HH, HT, TH, TT\}$$

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$S_C = \{HHH, \text{HHT}, \text{H}TH, \text{H}TT, \text{T}HH, \text{T}HT, \text{T}TH, TTT\}$

$S_D = \{\text{Spades, Diamonds, Hearts, Clubs}\}$

Illustration 2

A coin is tossed. If it shows head, we draw a ball from a bag consisting of 3 blue and 4 white balls; if it shows tail we throw a die. Describe the sample space of this experiment.

Solution:

Let us denote blue balls by B_1, B_2, B_3 and the white balls by W_1, W_2, W_3, W_4. Then a sample space of the experiment is

$S = \{HB_1, HB_2, HB_3, HW_1, HW_2, HW_3, HW_4, T1, T2, T3, T4, T5, T6\}$.

Here HB_i means head on the coin and ball B_i is drawn, HW_i means head on the coin and ball W_i is drawn. Similarly, Ti means tail on the coin and the number i on the die.

Illustration 3

Consider the experiment in which a coin is tossed repeatedly until a head comes up. Describe the sample space.

Solution:

In the experiment head may come up on the first toss, or the 2nd toss, or the 3rd toss and so on. Hence, the desired sample space is $S = \{H, TH, TTH, TTTH, TTTTH,...\}$

Illustration 4

A coin is tossed three times, consider the following events.

A : 'no head appears'

B : 'exactly one head appears'

C : 'at least two heads appear'

Do they form a set of mutually exclusive and exhaustive events ?

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Solution:

The sample space of the experiment is

$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

Events A, B and C are given by

$A = \{TTT\}$

$B = \{HTT, THT, TTH\}$

$C = \{HHT, HTH, THH, HHH\}$

Now,

$A \cup B \cup C = \{TTT, HTT, THT, TTH, HHT, HTH, THH, HHH\} = S$

Therefore A, B and C are exhaustive events. Also, $A \cap B = \phi$, $A \cap C = \phi$ and $B \cap C = \phi$. Therefore, the events are pair-wise disjoint, i.e., they are mutually exclusive. Hence, A, B and C form a set of mutually exclusive and exhaustive events.

3. CLASSICAL DEFINITION OF PROBABILITY:

If 'n' represents the total number of equally likely, mutually exclusive and exhaustive outcomes of a random experiment and 'm' of them are favourable to the happening of the event A, then the probability of happening of the event A is given by $P(A) = \frac{m}{n}$.

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Let E denote an experiment of rolling a fair die once. Let us define a few sample spaces associated with experiment E.

$S_1 = \{1,2,3,4,5,6\}$

$S_2 = \{1 \text{ or } 2, 3 \text{ or } 5, 4, 6\}$

$S_3 = \{\text{prime, composite}\}$

$S_4 = \{\text{greater than 4, less than 4}\}$

Let event A be defined as 'outcome is a prime number'. Then probability of event A is

$$P(A) = \underbrace{\frac{3}{6}\Big|_{\text{using } S_1}}_{2,3,5 \text{ are prime}}, \underbrace{\frac{2}{4}\Big|_{\text{using } S_2}}_{1 \text{ or } 2,\ 3 \text{ or } 5}, \underbrace{\frac{1}{2}\Big|_{\text{using } S_3}}_{\text{prime}} \text{ and } \underbrace{\frac{2}{2}\Big|_{\text{using } S_4}}_{2,3<4<5}$$

The probability we get using S_2 is **wrong** because the elements of S_2 are **NOT equally likely.**

The probability we get using S_3 is **wrong** because the elements of S_2 are **neither equally likely nor exhaustive.**

The probability we get using S_4 is **wrong** because the elements of S_2 are **NOT mutually exclusive.**

Thus, its very important that the elements of sample spaced used to find probability using Classical Definition must be **equally likely, mutually exclusive and exhaustive.**

Note :

(i) $0 \le P(A) \le 1$

(ii) There are $(n-m)$ outcomes which are favourable to the event that A does not happen. 'The event A does not happen' is denoted by $\bar{A}$ (and is read as 'not A')

Thus $P(\bar{A}) = \frac{n-m}{n} = 1 - \frac{m}{n}$ i.e. $P(\bar{A}) = 1 - P(A)$

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3.1 ODDS:

Let A be an event. If x cases are favourable to A & y cases are favourable to, then we say that

Odds in favour of *A* are $x : y$ i.e. $\big(P(A)\big) : \big(1-P(A)\big)$

Odds against *A* are $y : x$ i.e. $\big(1-P(A)\big) : P(A)$

Illustration 5

If a 12 letter word is formed using the letters I,N,T,E,R,M,E,D,I,A,T,E are arranged, then find the odds in favour of the event that no two 'E's occur together.

Solution:

$I \to 2, N \to 1, T \to 2, E \to 3, R \to 1, M \to 1, D \to 1, A \to 1$ (3'E's, Rest 9 letters)

First arrange rest of the letters = $\frac{9!}{2!\ 2!}$,

Now 3'E's can be placed by ${}^{10}C_3$ ways, so favourable cases = $\frac{9!}{2!\ 2!} \times {}^{10}C_3 = 3 \times 10!$

Total cases = $\frac{12!}{2!\,2!\,3!} = \frac{11}{2} \times 10!$; Non-favourable cases = $\left(\frac{11}{2} - 3\right) \times 10! = \frac{5}{2} \times 10!$

Odds in favour of the event = $\frac{3}{5/2} = \frac{6}{5}$

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Illustration 6

From a group of 10 persons consisting of 5 lawyers, 3 doctors and 2 engineers, four persons are selected at random. Find the probability that the selection contains at least one person of each profession.

Solution:

$n(S) = {}^{10}C_4 = 210.$

$n(E) = {}^5C_2 \times {}^3C_1 \times {}^2C_1 + {}^5C_1 \times {}^3C_2 \times {}^2C_1 + {}^5C_1 \times {}^3C_1 \times {}^2C_2 = 105$

$\therefore\ P(E) = \dfrac{105}{210} = \dfrac{1}{2}.$

Illustration 7

If four cards are drawn at random from a well-shuffled pack of playing cards, then find the probability that at least one of them is an ace.

Solution:

If A is a combination of four cards containing at least one ace (i.e. either one ace, or two aces, or three aces or four aces) then $\bar{A}$ is a combination of four cards containing no aces.

Now $P(\bar{A}) = \dfrac{\text{Number of combinations of four cards with no aces}}{\text{Total number of combinations of four cards}} = \dfrac{{}^{48}C_4}{{}^{52}C_4} = 0.72$

Using $P(A) + P(\bar{A}) = 1$ we have $P(A) = 1 - P(\bar{A}) = 1 - 0.72 = 0.28$

Illustration 8

A bag contains n white and n red balls. Pairs of balls are drawn without replacement until the bag is empty. Find the probability that each pair consists of one white and one red ball.

Solution:

Let S be the sample space & E be the event that each of the n pairs of balls drawn consists of one white and one red ball.

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$\therefore\ n(S) = ({}^{2n}C_2)\ ({}^{2n-2}C_2)\ ({}^{2n-4}C_2)...({}^4C_2)({}^2C_2)$

$= \left\{\dfrac{(2n)(2n-1)}{1.2}\right\}\left\{\dfrac{(2n-2)(2n-3)}{1.2}\right\}\left\{\dfrac{(2n-4)(2n-5)}{1.2}\right\}.....\left\{\dfrac{4.3}{1.2}\right\}.\left\{\dfrac{2.1}{1.2}\right\}$

$= \dfrac{1.2.3.4....(2n-1)2n}{2^n} = \dfrac{2n!}{2^n}$

and $n(E) = ({}^nC_1.{}^nC_1)\ ({}^{n-1}C_1.{}^{n-1}C_1)\ ({}^{n-2}C_1.{}^{n-2}C_1)...({}^2C_1.{}^2C_1)\ ({}^1C_1.{}^1C_1)$

$= n^2.(n-1)^2.(n-2)^2....2^2.1^2 = [1.2.3.....(n-1)n]^2 = (n!)^2$

$\therefore$ Required Probability, $P(E) = \dfrac{n(E)}{n(S)} = \dfrac{(n!)^2}{(2n)!/2^n} = \dfrac{2^n}{\dfrac{2n!}{(n!)^2}} = \dfrac{2^n}{{}^{2n}C_n}.$

Illustration 9

5 different marbles are placed in 5 different boxes randomly. Find the probability that exactly two boxes remain empty. Given each box can hold any number of marbles

Solution:

Total ways to place 5 different marbles in 5 different boxes $= 5^5$.

Number of ways to place these marbles such that exactly two of the boxes remains empty

$= {}^5C_3\left(3^5 - {}^3C_1(2)^5 + {}^3C_2(1)^5 - {}^3C_3(0)^5\right)$

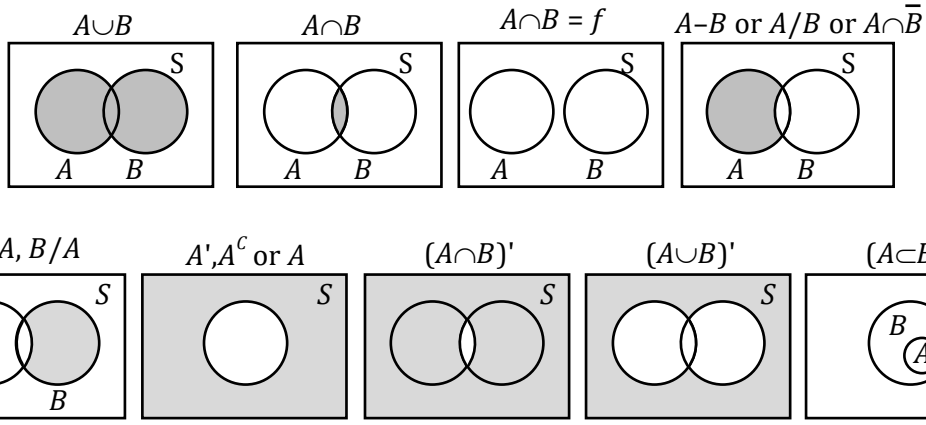
Hence, required Probability $= \dfrac{(10)(150)}{5^5} = \dfrac{24}{25}.$

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4. VENN DIAGRAMS:

A diagram used to illustrate relationships between sets. Commonly, a rectangle represents the universal set and any closed figure within it represents a given set (all members of the given set are represented by points within the closed figure).

Let S is the sample space of an experiment and A, B are two events corresponding to it :

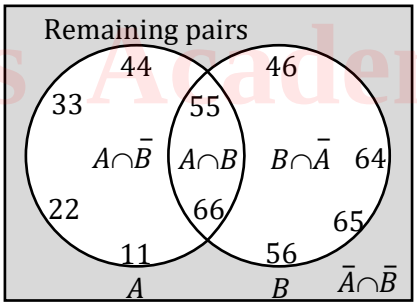


Example : Let us conduct an experiment of tossing a pair of dice.

Two events defined on the experiment are

A : getting a doublet 11, 22, 33, 44, 55, 66}

B : getting total score of 10 or more {64, 46, 55, 56, 65, 66}



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5. DE MORGAN'S LAW:

If A & B are two subsets of a universal set U, then

(i) $(A \cup B)^c = A^c \cap B^c$ & (ii) $(A \cap B)^c = A^c \cup B^c$

Note :

(a) $(A \cup B \cup C)^C = A^C \cap B^C \cap C^C$ & $(A \cap B \cap C)^C = A^C \cup B^C \cup C^C$

(b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ & $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

6. ADDITION THEOREM :

6.1 FOR TWO EVENTS:

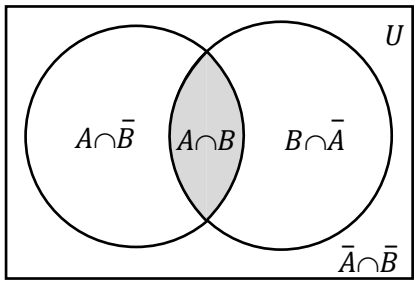
Let A and B are any two events.

Then according to Addition Theorem, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Also, note the following points for events A and B:

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(a)
$$\begin{bmatrix} P(A\cup B) \\ P(A+B) \\ P(A \text{ or } B) \\ P(\text{occurence of atleast one of } A \text{ or } B) \\ P(\text{either } A \text{ or } B) \end{bmatrix} = \begin{bmatrix} P(A)+P(B)-P(A\cap B) \\ P(A)+P(\bar{A}\cap B) \\ P(B)+P(A\cap\bar{B}) \\ P(A\cap B)+P(\bar{A}\cap B)+P(A\cap\bar{B}) \\ 1-P(\overline{A\cup B}) \\ 1-P(\bar{A}\cap\bar{B}) \end{bmatrix}$$

Note :

(i) If A & B are mutually exclusive then $P(A\cup B) = P(A) + P(B)$.

(ii) If A & B are mutually exclusive and exhaustive, then $P(A\cup B) = P(A) + P(B) = 1$.

(b) $P(\text{only } A \text{ occurs}) = P(A\backslash B) = P(A - B) = P(A \cap B^c) = P(A) - P(A \cap B)$

(c) $P(\text{exactly one of } A, B \text{ occurs}) = P(A\cap\bar{B}) + P(B\cap\bar{A})$

$$= \boldsymbol{P(A) + P(B) - 2P(A\cap B)}$$

$$= P(A\cup B) - P(A\cap B) = P(A^c\cup B^c) - P(A^c\cap B^c)$$

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(d) P(none of A, B occurs) $= P(\bar{A}\cap\bar{B}) = P(\overline{A\cup B})$

$$= 1-P(A)-P(B)+P(A\cap B)$$

(e) P(at most one of A, B occurs) $= 1-P(A\cap B)$

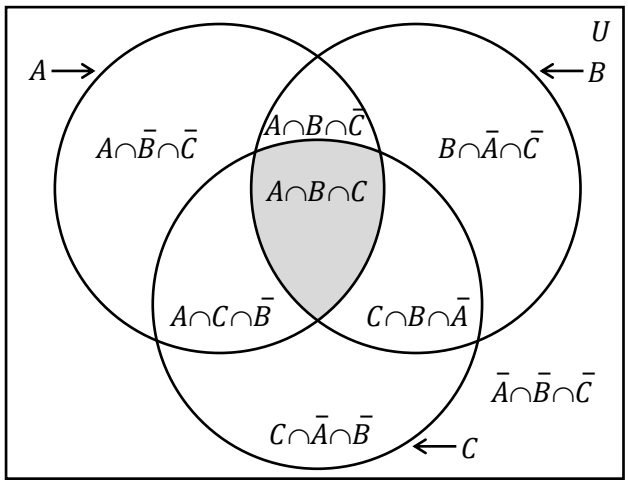
$$= P(A\cap\bar{B})+P(\bar{A}\cap B)+P(\bar{A}\cap\bar{B})$$

(f) $\max(0, P(A)+P(B)-1) \le P(A\cap B) \le \min(P(A), P(B))$

(g) $\max(P(A), P(B)) \le P(A\cap B) \le \min(1, P(A)+P(B))$

6.2 FOR THREE EVENTS :

Let A, B and C are any two events.



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Then according to Addition Theorem, we have

$\boldsymbol{P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)}$

Also, note the following points:

(a) P(at least one of A, B or C occurs)

$= P(A \cup B \cup C)$

$= P(A)+P(B)+P(C)-P(A \cap B)-P(B \cap C)-P(C \cap A)+P(A \cap B \cap C)$

(b) P(only A occurs)

$= P(A \cap \bar{B} \cap \bar{C})$

$= P(A)-P(A \cap B)-P(A \cap C)+P(B \cap C)$

(c) P(exactly one of A, B, C occurs)

$= P$(only A occurs) + P(only B occurs) + P(only C occurs)

$= P(A \cap \bar{B} \cap \bar{C})+P(\bar{A} \cap B \cap \bar{C})+P(\bar{A} \cap \bar{B} \cap C)$

$= P(A \cup B \cup C)-P(A \cap B)-P(B \cap C)-P(C \cap A)+2P(A \cap B \cap C)$

$= P(A)+P(B)+P(C)-2P(A \cap B)-2P(B \cap C)-2P(C \cap A)+3P(A \cap B \cap C)$

(d) P(exactly two of A, B, C occurs)

$= P$(only A and B occurs) + P(only B and C occurs) + P(only C and A occurs)

$= P(A \cap B \cap \bar{C})+P(\bar{A} \cap B \cap C)+P(A \cap \bar{B} \cap C)$

$= P(A \cap B)+P(B \cap C)+P(C \cap A)-3P(A \cap B \cap C)$

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(e) P(at least two of A, B, C occurs)

$= P$(only A and B occurs) + P(only B and C occurs)

+ P(only C and A occurs) + P(all of A, B, C occurs)

$= P(A \cap B \cap \bar{C})+P(\bar{A} \cap B \cap C)+P(A \cap \bar{B} \cap C)+P(A \cap B \cap C)$

$= P(A \cap B)+P(B \cap C)+P(C \cap A)-2P(A \cap B \cap C)$

(f) P(at most one of A, B, C occurs)

$= P$(none of A, B, C occurs) + P(only A occurs) + P(only B occurs) + P(only C occurs)

$= P(\bar{A} \cap \bar{B} \cap \bar{C})+P(A \cap \bar{B} \cap \bar{C})+P(\bar{A} \cap B \cap \bar{C})+P(\bar{A} \cap \bar{B} \cap C)$

$= 1-P(\text{at least 2 of } A, B, C \text{ occurs})$

$= 1-P(A \cap B)-P(B \cap C)-P(C \cap A)+2P(A \cap B \cap C)$

(g) P(at most two of A, B, C occurs)

$= 1-P(\text{all of } A, B, C \text{ occurs})$

$= 1-P(A \cap B \cap C)$

(h) P(none of A, B, C occurs)

$= P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\overline{A \cup B \cup C})$

$= 1-P(A \cup B \cup C)$

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Illustration 10

Given two events A and B. If odds against A are as 2 : 1 and those in favour of $A \cup B$ are as 3 : 1, then find the range of $P(B)$.

Solution:

Clearly $P(A) = 1/3$, $P(A \cup B) = 3/4$.

Now, $P(B) \le P(A \cup B)$

$\Rightarrow$ $P(B) \le 3/4$

Also, $P(B) = P(A \cup B) - P(A) + P(A \cap B)$

$\Rightarrow$ $P(B) \ge P(A \cup B) - P(A)$ $(\because P(A \cap B) > 0)$

$\Rightarrow$ $P(B) \ge 3/4 - 1/3$ $\Rightarrow$ $P(B) \ge \frac{5}{12}$

$\Rightarrow$ $\frac{5}{12} \le P(B) \le \frac{3}{4}$

Illustration 11

If A and B are two events such that $P(A \cup B) = \frac{3}{4}$, $P(A \cap B) = \frac{1}{4}$ and $P(A^c) = \frac{2}{3}$. Then find -

(i) $P(A)$ (ii) $P(B)$ (iii) $P(A \cap B^c)$ (iv) $P(A^c \cap B)$

Solution:

$P(A) = 1 - P(A^c) = 1 - \frac{2}{3} = \frac{1}{3}$

$P(B) = P(A \cup B) + P(A \cap B) - P(A) = \frac{3}{4} + \frac{1}{4} - \frac{1}{3} = \frac{2}{3}$

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$P(A \cap B^c) = P(A) - P(A \cap B) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

$P(A^c \cap B) = P(B) - P(A \cap B) = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$

Illustration 12

The probability that a person will get an electric contract is 2/5 and the probability that he will not get plumbing contract is 4/7. If the probability of getting at least one contract is 2/3, what is the probability that he will get both?

Solution:

Let A denote the events that the person will get an electric contract, and B A denote the events that the person will get an electric contract.Then

$P(A) = \frac{2}{5}, P(\bar{B}) = \frac{4}{7}$ $P(A \cup B) = \frac{2}{3}$.

$P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$\Rightarrow P(A \cap B) = \frac{2}{5} + \frac{3}{7} - \frac{2}{3} = \frac{42 + 45 - 70}{105} = \frac{17}{105}$

Illustration 13

Let A,B,C be three events. If the probability of occurring exactly one event out of A and B is $1 - a$, out of B and C is $1 - 2a$, out of C and A is $1 - a$ and that of occurring three events simultaneously is a^2, then prove that the probability that at least one out of A,B,C will occur is greater than 1/2.

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Solution

$P(A) + P(B) - 2P(A \cap B) = 1 - a$ ...(1)

and $P(B) + P(C) - 2P(B \cap C) = 1 - 2a$ ...(2)

and $P(C) + P(A) - 2P(C \cap A) = 1 - a$ ...(3)

and $P(A \cap B \cap C) = a^2$ ...(4)

$\because P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

$= \frac{1}{2}\{P(A)+P(B)-2P(A\cap B)+P(B)+P(C)-2P(B\cap C)+P(C)+P(A)-2P(C\cap A)\}+P(A\cap B\cap C)$

$= \frac{1}{2}\{1 - a + 1 - 2a + 1 - a\} + a^2$ {from (1), (2), (3) & (4)}

$= \frac{3}{2} - 2a + a^2 = (a - 1)^2 + \frac{1}{2} > \frac{1}{2}$

7. CONDITIONAL PROBABILITY AND MULTIPLICATION THEOREM:

7.1 Conditional Probability:

Let A and B be two events such that $P(A) > 0$. Then conditional probability of B given that A has occurred is given by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Since A is known to have occurred, it becomes the new sample space replacing the original S

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7.2 Multiplication Theorem:

In probability multiplication theorem is defined as

$P(A\cap B)=P(A)\cdot P(B|A)=P(B)\cdot P(A|B)$

This is also called compound probability. It says the probability that both A and B occur is equal to the probability that A occur times the probability that B occurs given that A has occurred.

Note : For any three events A_1, A_2, A_3 we have

$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | (A_1 \cap A_2))$

Illustration 14

Two dice are thrown. Find the probability that the numbers appeared have a sum of 8 if it is known that the second die always exhibits 4

Solution:

Let A be the event of occurrence of 4 always on the second die

$=\{(1,4),(2,4),(3,4),(4,4),(5,4),(6,4)\}$; $\therefore\ n(A)=6$

and B be the event of occurrence of such numbers on both dice whose sum is 8 = {(6,2), (5,3), (4,4), (3,5), (2,6)}.

Thus, $A\cap B=A\cap\{(4,4)\}=\{(4,4)\}$

$\therefore\ n(A\cap B)=1$

$\therefore\ P(B/A)=\frac{n(A\cap B)}{n(A)}=\frac{1}{6}$ or $\frac{P(A\cap B)}{P(A)}=\frac{1/36}{6/36}=\frac{1}{6}$

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Illustration 15

A bag contains 3 red, 6 white and 7 blue balls. Two balls are drawn one by one. What is the probability that first ball is white and second ball is blue when first drawn ball is not replaced in the bag?

Solution:

Let A be the event of drawing first ball white and B be the event of drawing second ball blue.

Here A and B are dependent events.

$$P(A)=\frac{6}{16},\ P(B|A)=\frac{7}{15}$$

$$P(AB)=P(A).P(B|A)=\frac{6}{16}\times\frac{7}{15}=\frac{7}{40}$$

Illustration 16

A bag contains 4 red and 4 blue balls. Four balls are drawn one by one from the bag, then find the probability that the drawn balls are in alternate colour.

Solution:

E_1 : Event that first drawn ball is red, second is blue and so on.

E_2 : Event that first drawn ball is blue, second is red and so on.

$$\therefore\quad P(E_1)=\frac{4}{8}\times\frac{4}{7}\times\frac{3}{6}\times\frac{3}{5} \text{ and } P(E_2)=\frac{4}{8}\times\frac{4}{7}\times\frac{3}{6}\times\frac{3}{5}$$

$$P(E)=P(E_1)+P(E_2)=2\times\frac{4}{8}.\frac{4}{7}.\frac{3}{6}.\frac{3}{5}=\frac{6}{35}$$

Illustration 17

If two events A and B are such that $P(\bar{A})=0.3$, $P(B)$ = 0.4 and $P(A\bar{B})=0.5$ then find $P(B|(A\cup\bar{B}))$.

Solution:

We have $P(B|(A\cup\bar{B}))=\frac{P[B\cap(A\cup\bar{B})]}{P(A\cup\bar{B})}=\frac{P[(B\cap A)\cup(B\cap\bar{B})]}{P(A)+P(\bar{B})-P(A\cap\bar{B})}$

$$=\frac{P(AB)}{P(A)+P(\bar{B})-P(A\bar{B})}=\frac{P(A)-P(A\bar{B})}{P(A)+P(\bar{B})-P(A\bar{B})}=\frac{0.7-0.5}{0.7+0.6-0.5}=\frac{0.2}{0.8}=\frac{1}{4}.$$

Illustration 18

Three coins are tossed. Two of them are fair and one is biased so that a head is three times as likely as a tail. Find the probability of getting two heads and a tail.

Solution:

E_1 : Event that head occurs on fair coin

E_2 : Event that, head occurs on biased coin

$$\therefore\ P(E_1)=\frac{1}{2},\ P(E_2)=\frac{3}{4}$$

E : HHT or HTH or THH

$$\Rightarrow P(E)=\frac{1}{2}\times\frac{1}{2}\times\frac{1}{4}+\frac{1}{2}\times\frac{1}{2}\times\frac{3}{4}+\frac{1}{2}\times\frac{1}{2}\times\frac{3}{4}=\frac{7}{16}$$

Illustration 19

In a multiple choice test of three questions there are five alternative answers given to the first two questions each and four alternative answers given to the last question. If a candidate guesses answers at random, what is the probability that he will get-

(a) Exactly one right answer ? (b) At least one right answer ?

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Solution:

E_1 : Event that, candidate guesses a correct answer for I question

E_2 : Event that, candidate guesses a correct answer for II question

E_3 : Event that, candidate guesses a correct answer for III question

$\therefore\ P(E_1)=\frac{1}{5},\ P(E_2)=\frac{1}{5},\ P(E_3)=\frac{1}{4}$

(a) E : Event that candidate get exactly one correct answer.

$\therefore\ P(E)=P(E_1)P(\bar{E}_2)P(\bar{E}_3)+P(\bar{E}_1).P(E_2)P(\bar{E}_3)+P(\bar{E}_1)P(\bar{E}_2)P(E_3)$

$=\frac{1}{5}\cdot\frac{4}{5}\cdot\frac{3}{4}+\frac{4}{5}\cdot\frac{1}{5}\cdot\frac{3}{4}+\frac{4}{5}\cdot\frac{4}{5}\cdot\frac{1}{4}=\frac{2}{5}$

(b) E : Event that candidate gets atleast one correct answer

$\therefore\ P(E)=1-P(\bar{E}_1)P(\bar{E}_2)P(\bar{E}_3)=1-\frac{4}{5}.\frac{4}{5}.\frac{3}{4}=\frac{13}{25}$

Illustration 20

A speaks truth in 75% cases and B in 80% cases. What is the probability that they contradict each other in stating the same fact?

Solution:

There are two mutually exclusive cases in which they contradict each other i.e. $\bar{A}B$ and $A\bar{B}$. Hence required probability = $P(A\bar{B}+\bar{A}B)=P(A\bar{B})+P(\bar{A}B)$

$= P(A)P(\bar{B})+P(\bar{A})P(B) = \frac{3}{4}.\frac{1}{5}+\frac{1}{4}.\frac{4}{5}=\frac{7}{20}$

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8. INDEPENDENT EVENTS :

Two events A & B are said to be independent if occurrence or non occurrence of one does not affect the probability of the occurrence or non occurrence of other. If A and B are two independent events, then

$$P(A\cap B)=P(A)\cdot P(B)$$

Often this is taken as the definition of independent events.

(a) If the occurrence of one event affects the probability of the occurrence of the other event then the events are said to be **Dependent** or **Contingent.**

(b) If A and B are independent events, then

(i) events $\bar{A}$ and $\bar{B}$ are also independent i.e $P(\bar{A}\cap\bar{B})=P(\bar{A})\cdot P(\bar{B})$.

(ii) events A and $\bar{B}$ are also independent i.e $P(A\cap\bar{B})=P(A)\cdot P(\bar{B})$.

(iii) events $\bar{A}$ and B are also independent i.e $P(\bar{A}\cap B)=P(\bar{A})\cdot P(B)$.

(c) Three events A, B & C are independent if & only if **all** the following conditions hold ;

(i) $P(A \cap B) = P(A) . P(B)$ (ii) $P(B \cap C) = P(B). P(C)$

(iii) $P(C \cap A) = P(C) . P(A)$ and (iv) $P(A \cap B \cap C) = P(A) . P(B) . P(C)$

i.e. they must be pairwise as well as mutually independent.

(d) If three events A, B and C are pair wise mutually exclusive then they must be mutually exclusive.

i.e. $P(A \cap B) = P(B \cap C) = P(C \cap A) = 0 \Rightarrow P(A \cap B \cap C) = 0$.

However the converse of this may not true.

(e) Independent events are not in general mutually exclusive & vice versa.

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Illustration 21

If A and B are independent events such that $P(A\cap\bar{B})=\frac{1}{3}$ and $P(A\cup B)=\frac{11}{15}$, then find $P(A\cap B)$.

Solution:

$P(A) - P(A\cap B) = \frac{1}{3}$ and $P(A\cup B) = P(A) + P(B) - P(A\cap B) = \frac{11}{15}$

$\Rightarrow P(B)=\frac{6}{15}=\frac{2}{5}$ $\Rightarrow$ $P(A) - P(A)\,P(B) = \frac{1}{3}$ $\Rightarrow P(A) = \frac{5}{9}$

$\Rightarrow P(A\cap B) = P(A)\,P(B) = \frac{2}{5}\times\frac{5}{9}=\frac{2}{9}$

Illustration 22

A lot contains 50 defective and 50 non-defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as :

A = {The first bulb is defective}

B = {The second bulb is non-defective}

C = {The two bulbs are both defective or both non-defective}

Determine whether

(i) A, B, C are pairwise independent, (ii) A, B, C are independent.

Solution:

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We have $P(A) = \frac{50}{100}.1=\frac{1}{2}$; $P(B)=1.\frac{50}{100}=\frac{1}{2}$; $P(C)=\frac{50}{100}.\frac{50}{100}+\frac{50}{100}.\frac{50}{100}=\frac{1}{2}$

$A\cap B$ is the event that first bulb is defective and second is non-defective.

$\therefore\quad P(A\cap B) = \frac{1}{2}.\frac{1}{2}=\frac{1}{4}$

$A\cap C$ is the event that both bulbs are defective.

$\therefore\quad P(A\cap C) = \frac{1}{2}.\frac{1}{2}=\frac{1}{4}$

Similarly $P(B\cap C) = \frac{1}{4}$

Thus we have $P(A\cap B) = P(A)\,.\,P(B)$; $P(A\cap C) = P(A)\,.\,P(C)$; $P(B\cap C) = P(B).P(C)$

$\therefore$ A, B and C are pairwise independent.

There is no element in $A\cap B\cap C$

$\therefore\quad P(A\cap B\cap C) = 0$

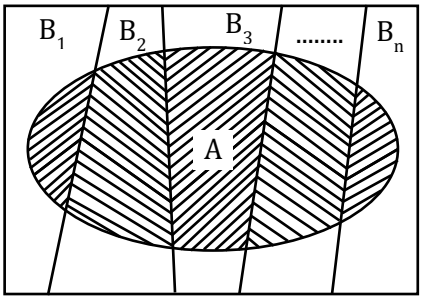
$\therefore\quad P(A\cap B\cap C)\neq P(A)\,.\,P(B)\,.\,P(C)$

Hence A, B and C are not mutually independent.

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9. TOTAL PROBABILITY THEOREM :

Let an event A of an experiment occurs with its n mutually exclusive & exhaustive events $B_1, B_2, B_3, \ldots\ldots B_n$ then total probability of occurrence of even A is



$$P(A)=P(AB_1)+P(AB_2)+P(AB_3)+\ldots+P(AB_n)$$
$$=P(B_1)\cdot P(A|B_1)+P(B_2)\cdot P(A|B_2)+P(B_3)\cdot P(A|B_3)+\ldots+P(B_n)\cdot P(A|B_n)$$
$$=\sum_{i=1}^{i=n} P(B_i)\cdot P(A|B_i)$$

Illustration 23

A purse contains 4 copper and 3 silver coins and another purse contains 6 copper and 2 silver coins. One coin is drawn from any one of these two purses. The probability that it is a copper coin is -

(A) $\frac{4}{7}$ (B) $\frac{3}{4}$ (C) $\frac{2}{7}$ (D) $\frac{37}{56}$

Ans. (D)

Solution:

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Let $A \equiv$ event of selecting first purse

$B \equiv$ event of selecting second purse

$C \equiv$ event of drawing a copper coin

Then given event has two disjoint cases: AC and BC

$$\therefore P(C)=P(AC+BC)=P(AC)+P(BC)=P(A)P(C|A)+P(B)P(C|B)$$

$$=\frac{1}{2}\cdot\frac{4}{7}+\frac{1}{2}\cdot\frac{6}{8}=\frac{37}{56}$$

Illustration 24

Three groups A, B, C are contesting for positions on the Board of Directors of a Company. The probabilities of their winning are 0.5, 0.3, 0.2 respectively. If the group A wins, the probability of introducing a new product is 0.7 and the corresponding probabilities for group B and C are 0.6 and 0.5 respectively. Find the probability that the new product will be introduced.

Solution:

Given $P(A) = 0.5$, $P(B) = 0.3$ and $P(C) = 0.2$

$P(A) + P(B) + P(C) = 1$

then events A, B, C are exhaustive.

If $P(E)$ = Probability of introducing a new product, then as given

$P(E|A) = 0.7$, $P(E|B) = 0.6$ and $P(E|C) = 0.5$

$P(E) = P(A).\ P(E|A) + P(B).\ P(E|B) + P(C).\ P(E|C)$

$= 0.5 \times 0.7 + 0.3 \times 0.6 + 0.2 \times 0.5 = 0.35 + 0.18 + 0.10 = 0.63$

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Illustration 25

A pair of dice is rolled together till a sum of either 5 or 7 is obtained. Find the probability that 5 comes before 7.

Solution:

Let E_1 = the event of getting 5 in a roll of two dice = {(1, 4), (2, 3), (3, 2), (4, 1)}

$\therefore\ P(E_1)=\frac{n(E_1)}{n(S)}=\frac{4}{6\times6}=\frac{1}{9}$

Let E_2 = the event of getting either 5 or 7

= {(1, 4), (2, 3), (3, 2), (4, 1), (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}

$\therefore\ P(E_2)=\frac{n(E_2)}{n(S)}=\frac{10}{6\times6}=\frac{5}{18}$

$\therefore$ the probability of getting neither 5 nor 7 $=P(\bar{E}_2)=1-P(E_2)=1-\frac{5}{18}=\frac{13}{18}$

The event of getting 5 before 7 = $E_1\cup(\bar{E}_2E_1)\cup(\bar{E}_2\bar{E}_2E_1)\cup\ldots\ldots$ to ∞

$\therefore$ the probability of getting 5 before 7

$=P(E_1)+P(\bar{E}_2E_1)+P(\bar{E}_2\bar{E}_2E_1)+\ldots.$ *to* $\infty=P(E_1)+P(\bar{E}_2)P(E_1)+P(\bar{E}_2)P(\bar{E}_2)P(E_1)+\ldots.$ *to* ∞

$=\frac{1}{9}+\frac{13}{18}.\frac{1}{9}+\frac{13}{18}.\frac{13}{18}.\frac{1}{9}+\ldots\ldots$ to ∞

$=\frac{1}{9}\left[1+\frac{13}{18}+\left(\frac{13}{18}\right)^2+\ldots\ldots\ to\ \infty\right]=\frac{1}{9}.\frac{1}{1-\frac{13}{18}}=\frac{1}{9}.\frac{18}{5}=\frac{2}{5}$

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10. BINOMIAL PROBABILITY DISTRIBUTION:

Suppose that we have an experiment such as tossing a coin or die repeatedly or choosing a marble from an urn repeatedly. Each toss or selection is called a trial. In any single trial there will be a probability associated with a particular event such as head on the coin, 4 on the die, or selection of a red marble. In some cases this probability will not change from one trial to the next (as in tossing a coin or die). Such trials are then said to be independent and are often called Bernoulli trials after James Bernoulli who investigated them at the end of the seventeenth century.

Let p be the probability that an event will happen in any single Bernoulli trial (called the probability of success). Then $q = 1 - p$ is the probability that the event will fail to happen in any single trial (called the probability of failure). The probability that the event will happen exactly x times in n trials (i.e., x successes and $n - x$ failures will occur) is given by the probability function.

$$f(x) = P(X = x) = \binom{n}{x}p^xq^{n-x}=\frac{n!}{x!(n-x)!}p^xq^{n-x} \qquad \ldots(i)$$

where the random variable X denotes the number of successes in n trials and $x = 0, 1, \ldots\ldots\ldots, n$.

Example : The probability of getting exactly 2 heads in 6 tosses of a fair coin is

$$P(X=2)=\binom{6}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{6-2}=\frac{6!}{2!4!}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{6-2}=\frac{15}{64}$$

The discrete probability function (i) is often called the binomial distribution since for $x = 0, 1, 2,\ldots,n$, it corresponds to successive terms in the binomial expansion

$$(q+p)^n=q^n+\binom{n}{1}q^{n-1}p+\binom{n}{2}q^{n-2}p^2+\ldots..+p^n=\sum_{x=0}^{n}\binom{n}{x}p^xq^{n-x}$$

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Illustration 26

If a fair coin is tossed 10 times, find the probability of getting

(i) exactly six heads (ii) atleast six heads (iii) atmost six heads

Solution

The repeated tosses of a coin are Bernoulli trials. Let X denotes the number of heads in an experiment of 10 trials.

Clearly, X has the binomial distribution with $n = 10$ and $p = \frac{1}{2}$

Therefore $P(X = x) = {}^{n}C_x q^{n-x} p^x$, $x = 0,1,2,....,n$

Here $n = 10$, $p = \frac{1}{2}$, $q = 1 - p = \frac{1}{2}$

Therefore $P(X = x) = {}^{10}C_x \left(\frac{1}{2}\right)^{10-x} \left(\frac{1}{2}\right)^{x} = {}^{10}C_x \left(\frac{1}{2}\right)^{10}$

Now (i) $P(X = 6) = {}^{10}C_6 \left(\frac{1}{2}\right)^{10} = \frac{10!}{6! \times 4! 2^{10}} = \frac{105}{512}$

(ii) P(atleast six heads)

$= P(X \geq 6) = P(X = 6) + P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$

$= {}^{10}C_6 \left(\frac{1}{2}\right)^{10} + {}^{10}C_7 \left(\frac{1}{2}\right)^{10} + {}^{10}C_8 \left(\frac{1}{2}\right)^{10} + {}^{10}C_9 \left(\frac{1}{2}\right)^{10} + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10}$

$= \left[\left(\frac{10!}{6! \times 4!}\right) + \left(\frac{10!}{7! \times 3!}\right) + \left(\frac{10!}{8! \times 2!}\right) + \left(\frac{10!}{9! \times 1!}\right) + \left(\frac{10!}{10!}\right)\right] \frac{1}{2^{10}} = \frac{193}{512}$

(iii) P(at most six heads) = $P(X \leq 6)$

$= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6)$

$= \left(\frac{1}{2}\right)^{10} + {}^{10}C_1 \left(\frac{1}{2}\right)^{10} + {}^{10}C_2 \left(\frac{1}{2}\right)^{10} + {}^{10}C_3 \left(\frac{1}{2}\right)^{10} + {}^{10}C_4 \left(\frac{1}{2}\right)^{10} + {}^{10}C_5 \left(\frac{1}{2}\right)^{10} + {}^{10}C_6 \left(\frac{1}{2}\right)^{10}$

$= \frac{848}{1024} = \frac{53}{64}$

Illustration 27

India and Pakistan play a 5 match test series of hockey, the probability that India wins at least three matches is -

(A) $\frac{1}{2}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{5}{16}$

Ans. (A)

Solution

India win atleast three matches

${}^{5}C_3 \left(\frac{1}{2}\right)^{5} + {}^{5}C_4 \left(\frac{1}{2}\right)^{5} + {}^{5}C_5 \left(\frac{1}{2}\right)^{5} = \left(\frac{1}{2}\right)^{5} (16) = \frac{1}{2}$

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Illustration 28

A coin is tossed 7 times. Each time a man calls head. The probability that he wins the toss on more than three occasions is -

(A) $\frac{1}{4}$ (B) $\frac{5}{8}$ (C) $\frac{1}{2}$ (D) none of these

Ans. (C)

Solution:

The man has to win at least 4 times.

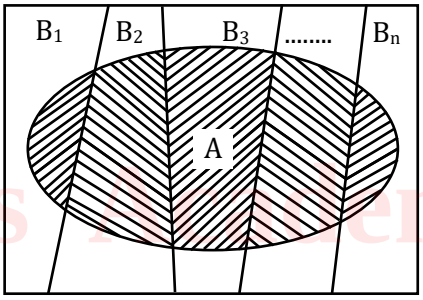
$\therefore$ the required probability

$$= {}^7C_4\left(\frac{1}{2}\right)^4 \cdot \left(\frac{1}{2}\right)^3 + {}^7C_5 \cdot \left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^2 + {}^7C_6\left(\frac{1}{2}\right)^6 \cdot \frac{1}{2} + {}^7C_7\left(\frac{1}{2}\right)^7$$

$$= ({}^7C_4 + {}^7C_5 + {}^7C_6 + {}^7C_7) \cdot \frac{1}{2^7} = \frac{64}{2^7} = \frac{1}{2}.$$

11. BAYE'S THEOREM :

Let $B_1, B_2, B_3,..., B_n$ be n mutually exclusive & exhaustive events. Let event A happens in conjunction with them and $P(A/B_1), P(A/B_2)..... P(A/B_n)$ are known. Then by Baye's Theorem, we have



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$$P(B_i \mid A) = \frac{P(B_i) \cdot P(A \mid B_i)}{\sum_{i=1}^{i=n} (P(B_i) \cdot P(A \mid B_i))}$$

Illustration 29

Given three identical boxes I, II and III, each containing two coins. In box I, both coins are gold coins, in box II, both are silver coins and in the box III, there is one gold and one silver coin. A person chooses a box at random and takes out a coin. If the coin is of gold, what is the probability that the other coin in the box is also of gold ?

Solution

Let E_1, E_2 and E_3 be the events that boxes I, II and III are chosen, respectively.

Then $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$

Also, let A be the event that 'the coin drawn is of gold'

Then $P(A|E_1)$ = P(a gold coin from box I) = $\frac{2}{2}$ = 1

$P(A|E_2)$ = P(a gold coin from box II) = 0

$P(A|E_3)$ = P(a gold coin from box III) = $\frac{1}{2}$

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Now, the probability that the other coin in the box is of gold
= the probability that gold coin is drawn from the box I.
= $P(E_1|A)$
By Baye's theorem, we know that

$$P(E_1|A)=\frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)+P(E_3)P(A|E_3)}=\frac{\frac{1}{3}\times 1}{\frac{1}{3}\times 1+\frac{1}{3}\times 0+\frac{1}{3}\times\frac{1}{2}}=\frac{2}{3}$$

Illustration 30

A bag A contains 2 white and 3 red balls and a bag B contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from the bag B.

Solution

Let E_1 = The event of ball being drawn from bag A
E_2 = The event of ball being drawn from bag B.
E = The event of ball being red.

Since, both the bags are equally likely to be selected, therefore

$$P(E_1)=P(E_2)=\frac{1}{2} \text{ and } P(E|E_1)=\frac{3}{5} \text{ and } P(E|E_2)=\frac{5}{9}$$

$\therefore$ Required probability

$$P(E_2|E)=\frac{P(E_2)P(E|E_2)}{P(E_1).P(E|E_1)+P(E_2)P(E|E_2)}=\frac{\frac{1}{2}\times\frac{5}{9}}{\frac{1}{2}\times\frac{3}{5}+\frac{1}{2}\times\frac{5}{9}}=\frac{25}{52}$$

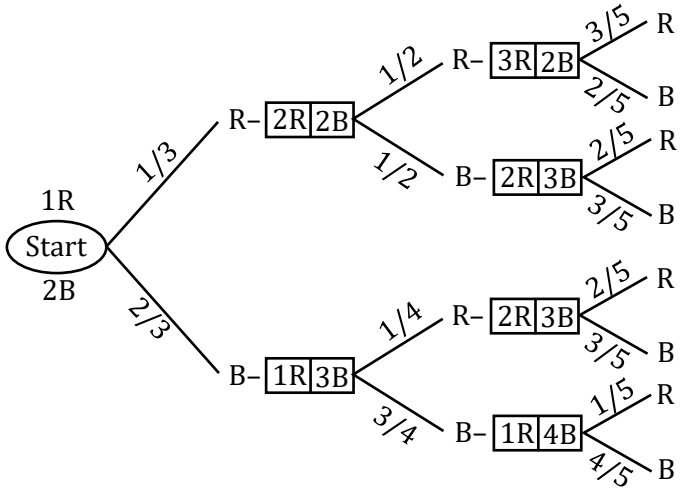
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12. PROBABILITY THROUGH STATISTICAL (STOCHATIC) TREE DIAGRAM :

These tree diagrams are generally drawn by economist and give a simple approach to solve a problem.

Illustration 31

A bag initially contains 1 red ball and 2 blue balls. A trial consists of selecting a ball at random, noting its colour and replacing it together with an additional ball of the same colour. Given that three trials are made, draw a tree diagram illustrating the various probabilities. Hence, or otherwise, find the probability that
(a) atleast one blue ball is drawn
(b) exactly one blue ball is drawn
(c) Given that all three balls drawn are of the same colour find the probability that they are all red.



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Solution :

Calculations :

$$P(A)=1-P(RRR)=1-\frac{1}{3}.\frac{1}{2}.\frac{3}{5}=1-\frac{1}{10}=\frac{9}{10}$$

$$P(\text{exactly one Blue})=\frac{1}{3}.\frac{1}{2}.\frac{2}{5}+\frac{1}{3}.\frac{1}{2}.\frac{2}{5}+\frac{2}{3}.\frac{1}{4}.\frac{2}{5}=\frac{1}{15}+\frac{1}{15}+\frac{1}{15}=\frac{3}{15}=\frac{1}{5}$$

$$P(C)=P\left(\frac{RRR}{(RRR\cup BBB)}\right)=\frac{P(RRR)}{P(RRR)+P(BBB)}=\frac{\frac{1}{3}.\frac{1}{2}.\frac{3}{5}}{\frac{1}{3}.\frac{1}{2}.\frac{3}{5}+\frac{2}{3}.\frac{3}{4}.\frac{4}{5}}=\frac{\frac{1}{10}}{\frac{1}{10}+\frac{4}{10}}=\frac{1}{5}$$

13. MISCELLANEOUS ILLUSTRATIONS :

Illustration 32

Three persons A, B, C in order cut a pack of cards, replacing them after each cut, on the condition that the first who cuts a spade shall win a prize; find their respective chances.

Solution

Let p be the chance of cutting a spade and q the chance of not cutting a spade from a pack of 52 cards.

Then $p=\frac{^{13}C_1}{^{52}C_1}=\frac{1}{4}$ and $q = 1 - \frac{1}{4}=\frac{3}{4}$

Now A will win a prize if he cuts spade at 1st, 4th, 7th, 10th turns, etc. Note that A will get a second chance if A, B, C all fail to cut a spade once and then A cuts a spade at the 4th turn. Similarly he will cut a spade at the 7th turn when A, B, C fail to cut spade twice, etc.

Hence A's chance of winning the prize

$$= p + q^3p + q^6p + q^9p + ...= \frac{p}{1-q^3}=\frac{\frac{1}{4}}{1-\left(\frac{3}{4}\right)^3}=\frac{16}{37}$$

Similarly B's chance = $(qp + q^4p + q^7p + ...) = q(p + q^3p + q^6p + ...) = \frac{3}{4}.\frac{16}{37}=\frac{12}{37}$

and C's chance = $\frac{3}{4}$ of B's chance = $\frac{3}{4}.\frac{12}{37}=\frac{9}{37}$

Illustration 33

(a) If p and q are chosen randomly from the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, with replacement, determine the probability that the roots of the equation $x^2 + px + q = 0$ are real.

(b) Each coefficient in the equation $ax^2 + bx + c = 0$ is determined by throwing an ordinary die. Find the probability that the equation will have equal roots.

Solution:

(a) If roots of $x^2 + px + q = 0$ are real, then $p^2 - 4q \geq 0$ .... (i)

Both p, q belongs to set S = {1, 2, 3, ... 10} when $p = 1$, no value of q from S will satisfy (i)

$p = 2$	$q = 1$ will satisfy	1 value
$p = 3$	$q = 1, 2$	2 value
$p = 4$	$q = 1, 2, 3, 4$	4 value
$p = 5$	$q = 1, 2, 3, 4, 5, 6$	6 value
$p = 6$	$q = 1, 2, 3, 4, 5, 6, 7, 8, 9$	9 value

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For p = 7, 8, 9, 10 all the ten values of q will satisfy.

Sum of these selections is 1 + 2 + 4 + 6 + 9 + 10 + 10 + 10 + 10 = 62

But the total number of selections of p and q without any order is 10 × 10 = 100

Hence the required probability is = $\frac{62}{100} = 0.62$

(b) Roots equal $\Rightarrow$ $b^2 - 4ac = 0$

$\therefore \left(\frac{b}{2}\right)^2 = ac$ ....(i)

Each coefficient is an integer, so we consider the following cases :

$b = 1$ $\therefore$ $\frac{1}{4} = ac$

No integral values of a and c

$b = 2$ $1 = ac$ $\therefore$ (1, 1)

$b = 3$ $9/2 = ac$

No integral values of a and c

$b = 4$ $4 = ac$ $\therefore$ (1, 4), (2, 2), (4, 1)

$b = 5$ $25/2 = ac$

No integral values of a and c

$b = 6$ $9 = ac$ $\therefore$ (3, 3)

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Thus we have 5 favourable ways for b = 2, 4, 6

Total number of equations is 6.6.6 = 216

$\therefore$ Required probability is $\frac{5}{216}$

Illustration 34

A set A has n elements. A subset P of A is selected at random. Returning the element of P, the set A is formed again and then a subset Q is selected from it. Find the probability that P and Q have no common elements.

Solution:

Let P be the empty set, or one element set or two elements set ..... or n elements set. Then the set Q will be chosen from amongst the remaining n elements or $n - 1$ elements or $n - 2$ elements ..... or no elements. The probability of P being an empty set is ${}^nC_0/2^n$, the probability of P being one element set is ${}^nC_1/2^n$ and in general, the probability of P being an r element set is ${}^nC_r/2^n$.

When the set P consisting of r elements is chosen from A, then the probability of choosing the set Q from amongst the remaining $n - r$ elements is $2^{n-r}/2^n$. Hence the probability that P and Q have no common elements is given by $\sum_{r=0}^{n} \frac{{}^nC_r}{2^n}.\frac{2^{n-r}}{2^n} = \frac{1}{4^n}\sum_{r=0}^{n} {}^nC_r.2^{n-r} = \left(\frac{1}{4}\right)^n (1+2)^n = \left(\frac{3}{4}\right)^n$ [By binomial theorem]

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Illustration 35

The probabilities of three events A, B and C are $P(A) = 0.6$, $P(B) = 0.4$ and $P(C) = 0.5$. If $P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$, $P(A \cap B \cap C) = 0.2$ and $P(A \cup B \cup C) \geq 0.85$, find $P(B \cap C)$.

Solution:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.8 = 0.6 + 0.4 - P(A \cap B)$

$\therefore \quad P(A \cap B) = 0.2$

Now $P(A \cup B \cup C) = S_1 - S_2 + S_3 = (0.6 + 0.4 + 0.5) - (0.2 + P(B \cap C) + 0.3) + 0.2 = 1.5 - 0.3 - P(B \cap C)$

We know $0.85 \leq P(A \cup B \cup C) \leq 1$

or $\quad 0.85 \leq 1.2 - P(B \cap C) \leq 1$

$\therefore \quad 0.2 \leq P(B \cap C) \leq 0.35$

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